ADRINE

Technical Assessment

Software Engineering | Product Thinking | UI/UX

Format	Practical Build
Time	3 Hours
Technology	Candidate's Choice
AI Tools	Allowed

01 The Assignment

Build a Clinic Management Application

Build a clinic management application for a modern outpatient clinic.

The product will be used by **reception staff and doctors** to manage patients and appointments.

The objective is not to build as many features as possible. We want to see how you **understand a problem, make technical and product decisions, build the solution, and polish the final experience.**

The finished application should feel like a product that could realistically be presented to a paying customer.

Core Workflow

Dashboard

Create a concise operational dashboard showing:

- Today's appointments
- Upcoming appointments
- Total patients
- Completed and cancelled appointments
- Relevant quick actions
- A useful appointment or activity visualisation

The dashboard should prioritise information that a clinic employee would actually need.

Patients

Create a patient management experience with:

- Patient list
- Search
- Filtering
- Sorting
- Pagination
- Add patient

- Edit patient
- Patient profile

The patient profile should include:

- Basic information
- Contact information
- Appointment history
- Clinical notes

Appointments

Create an appointment workflow that allows staff to:

- View appointments
- Filter by doctor, date and status
- Create appointments
- Reschedule appointments
- Cancel appointments
- Mark appointments as completed

Appointment conflicts or unavailable time slots should be clearly communicated in the interface.

Roles

Implement at least two roles.

Receptionist

- Manage patients
- Manage appointments

Doctor

- View their appointments
- View relevant patient information
- Add clinical notes

Users should not be able to access functionality outside of their permitted role.

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02 Product and Design

Design is part of the assessment.

The interface will be evaluated alongside the engineering.

The final product should demonstrate the visual discipline of a modern **B2B healthcare application**.

We will look at:

- Information hierarchy
- Typography
- Spacing and alignment
- Navigation
- Forms
- Tables
- Components

- Colour usage
- Loading states
- Empty states
- Validation states
- Error states
- Responsive behaviour
- Overall usability

Design expectation
The interface should feel **intentional and product-designed**, rather than assembled from a generic dashboard template.

Avoid:

- Excessive rounded cards
- Unnecessary gradients
- Random accent colours
- Excessive animations
- Visual clutter
- Generic AI-generated dashboard layouts
- Components that do not serve a clear purpose

A simple, consistent and well-designed interface is preferable to a visually complicated one.

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03 Engineering

The application must include a **real backend/API and persistent database**.

At minimum, model the following:

- User
- Doctor
- Patient
- Appointment

The frontend should communicate with the backend through APIs.

We expect

Frontend
Reusable components, sensible state management, responsive implementation and maintainable structure.

Backend
Clear API boundaries, validation, error handling and sensible separation of concerns.

Database
Sensible schema design, relationships, data integrity and appropriate querying.

Authentication
Authentication and role-based authorisation.

Reliability
Proper handling of loading, empty, error and invalid-input states.

Security
Reasonable handling of authentication, authorisation and user input.

04 Working Rules

You may use **any technology stack** you are comfortable with.

You may use:

- ChatGPT
- Claude
- Cursor
- GitHub Copilot
- Other coding agents
- Documentation
- Open-source libraries
- Frameworks
- Development tools

AI assistance is allowed.
However, you should understand and be able to explain the important parts of the implementation you submit.

We are evaluating your ability to **use tools effectively**, not your ability to write every line of code manually.

Do not over-engineer the solution. Good engineering judgement is more important than the number of technologies used.

05 Final Presentation

At the end of the assessment, you will demonstrate the application. Please be prepared to explain:

Product
Walk us through the main user journeys and demonstrate the application working.

Architecture
Explain how your frontend, backend and database communicate.

Data Model
Explain the important entities, relationships and decisions in your database.

Design
Explain 2–3 important UI/UX decisions you made and why.

Engineering
Show one part of the implementation that you consider particularly well designed.

Trade-offs
Explain what you deliberately did not build and why.

Next Steps
If you had another two days, what are the three highest-value improvements you would make?

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06 Live Engineering Test

After the main demonstration, we will introduce an additional requirement. You will modify your existing application **live**.

Requirement

> **A doctor may only view patients who have an appointment with that doctor.**

Implement this requirement across the relevant parts of the application without breaking the existing workflow. This may require changes to:

- Database queries
- API and backend logic
- Authorisation
- Frontend behaviour

Additional change
Add a patient status: **Active, Inactive or Archived**.

Then allow users to filter the patient list by status.

Target time: 10–15 minutes.

We are evaluating your understanding of the system, reasoning and ability to safely modify your own implementation, not typing speed.

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07 Engineering Judgement

Assume the clinic grows to:

- **100** doctors
- **1,000,000** patients
- **50,000** appointments per day

Explain:

1. Which parts of your current implementation would become bottlenecks?
2. What would you change first?
3. How would you scale the database?
4. How would you improve API performance?
5. What would you change on the frontend?

You do not need to implement the scaling solution.
We want to understand how you think about **systems beyond the initial prototype**.

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08 Evaluation

Area	Weight

Product judgement	15%
UI / UX	20%
Frontend engineering	15%
Backend engineering	15%
Database and data modelling	10%
Problem solving	15%
Code quality	10%
Total	**100%**

The Standard

We are **not looking for a perfect three-hour product**.

We are looking for evidence that you can:
Understand the problem → make good decisions → build cleanly → create a polished experience → explain your work → adapt when requirements change.

The quality of your **thinking, execution and judgement** matters more than the number of features you complete.

ADRINE

Building intelligent systems for healthcare.